

CE/CZ4171

Internet of Things:
Communications and Networking

IoT Case Study: Smart Grid



Outline

- **Introduction to Smart Grid**
- Smart Grid Communication Architecture
- Use Cases

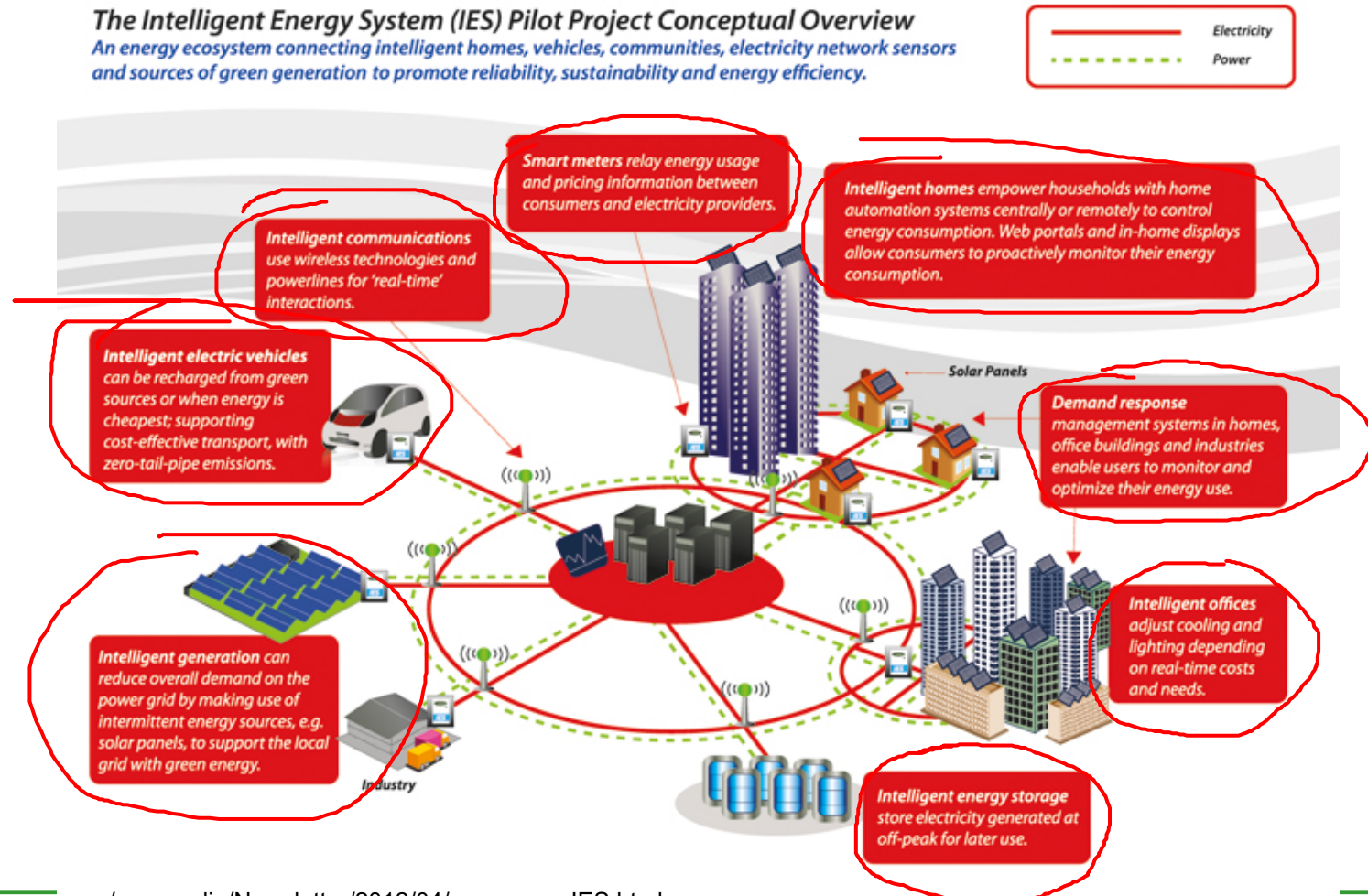


Introduction

- What is smart grid?
 - Smart grids – add communication capabilities and intelligence to traditional power grids

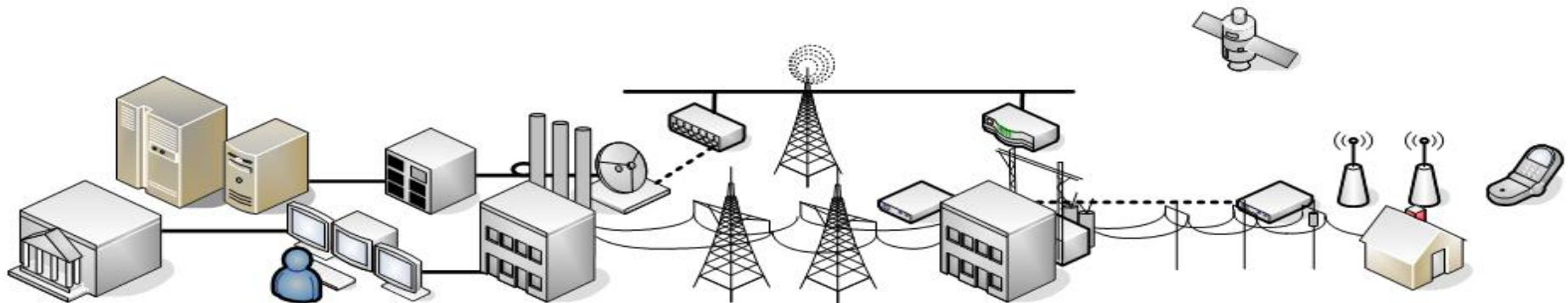
The Intelligent Energy System (IES) Pilot Project Conceptual Overview

An energy ecosystem connecting intelligent homes, vehicles, communities, electricity network sensors and sources of green generation to promote reliability, sustainability and energy efficiency.

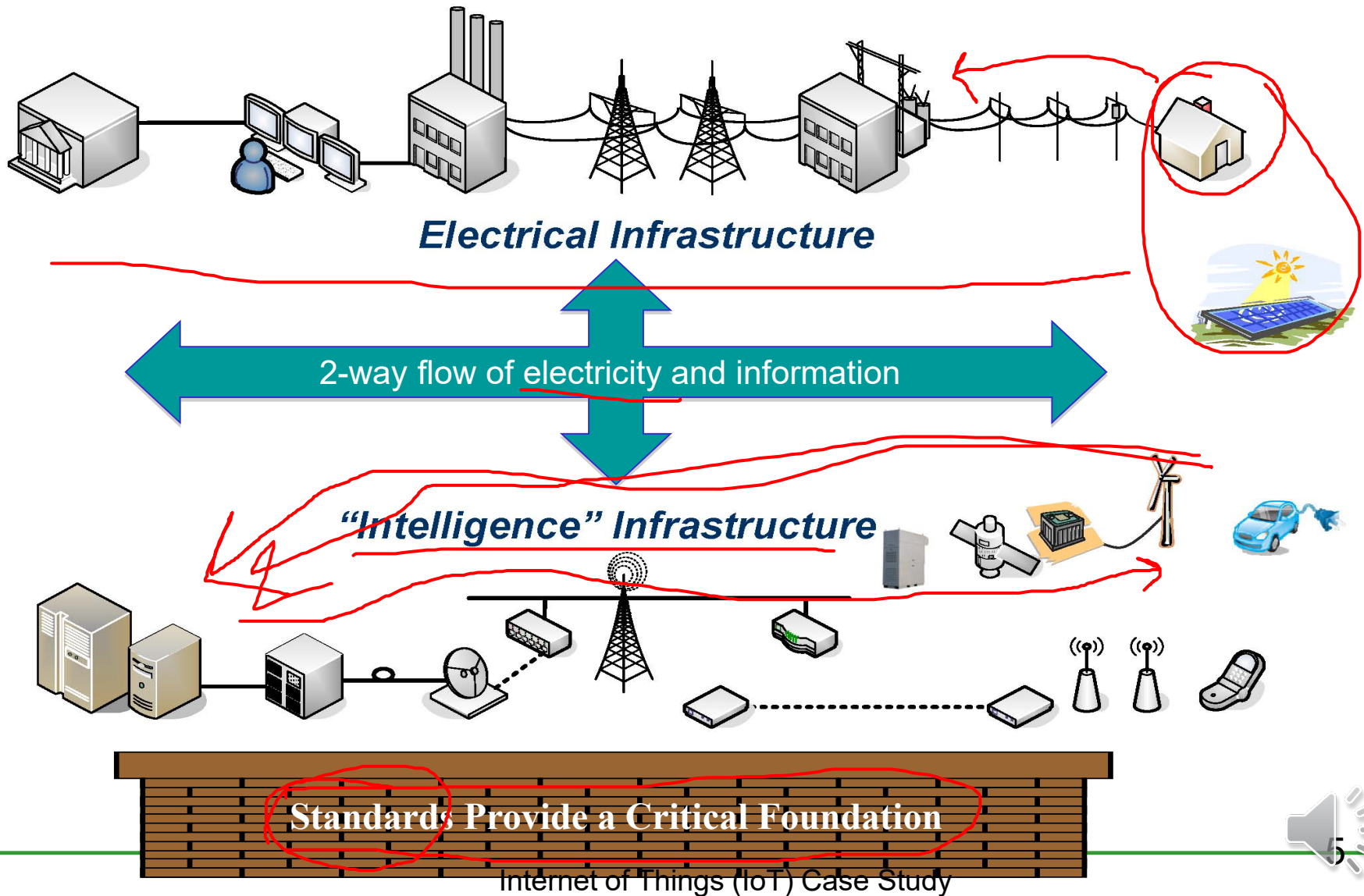


Introduction

- What enables smart grids?
 - Internet of Things (IoT): Intelligent sensors and actuators
 - Extended data management system
 - Expanded two way communications between power generation, distribution, and customers
 - Network security
 - Data analytics

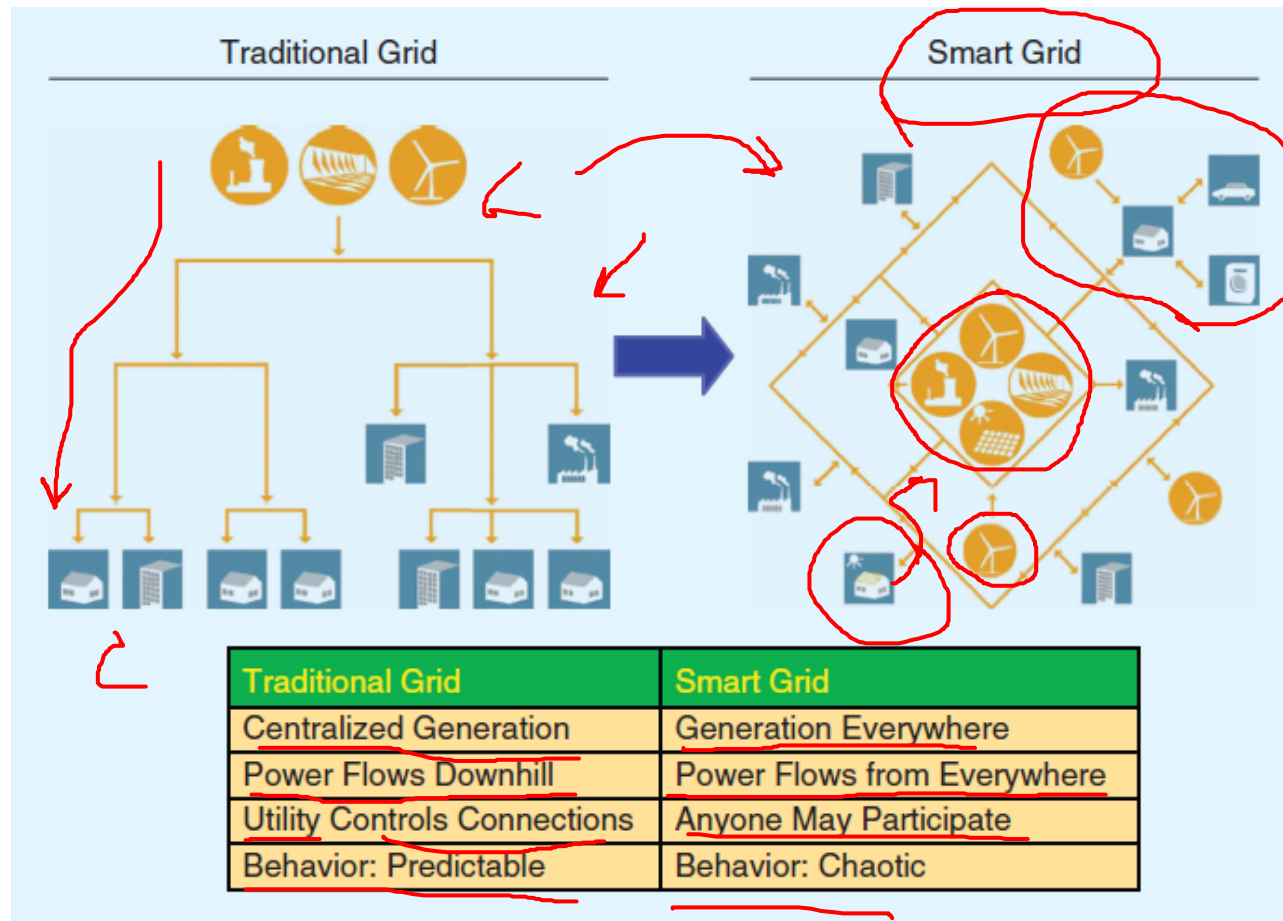


Smart Grid: The “Energy Internet”



Transition

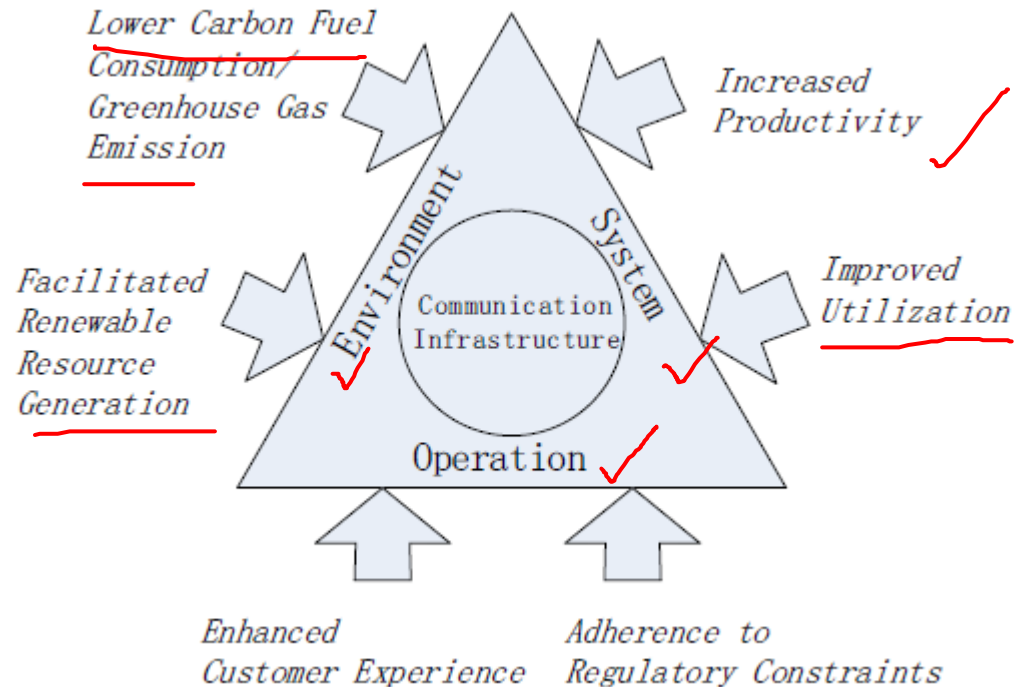
- Transition from traditional power grid to smart grid



Motivations

Smart Grid Enables:

- Higher Penetration of Renewables
- Smart Charging of Electric Vehicles
- Consumers to Control Energy Bills
- Efficient Grid Operations & Reduced Losses
- Reduced Distribution Outages
- Improved System Reliability & Security



Primary objectives

- National integration
- Self healing and adaptive –Improve distribution and transmission system operation
- Allow customers freedom to purchase power based on dynamic pricing
- Improved quality of power-less wastage
- Integration of large variety of generation options



Economic and Social Benefits

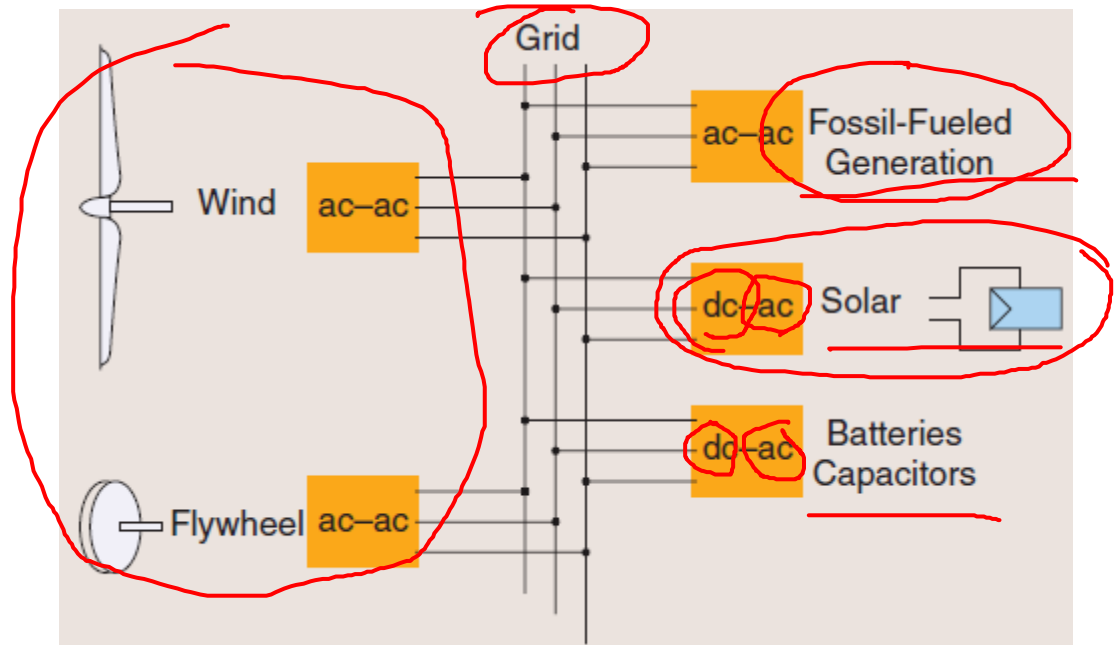
- Provide Customer Benefits
- Reduce Peak Demand
- Increase Energy Conservation & Efficiency
- Reduce Operating Expenses
- Increase Utility Worker Safety
- Improve Grid Resiliency and Reliability
- Reduce Greenhouse Gas Emissions
- Promote Energy Independence
- Promote Economic Growth & Productivity



Distributed Generation

- *Hybrid Energy Resource*

- Fossil-Fuel
- Wind
- Solar
- Bio-Mass
- Batteries
- Capacitors
- Flywheel
- Etc.



Smart Metering

- Automatic Metering
 - Automatic Meter Reading (AMR)
 - Automated Metering Management (AMM)
 - Advanced Metering Infrastructure (AMI)

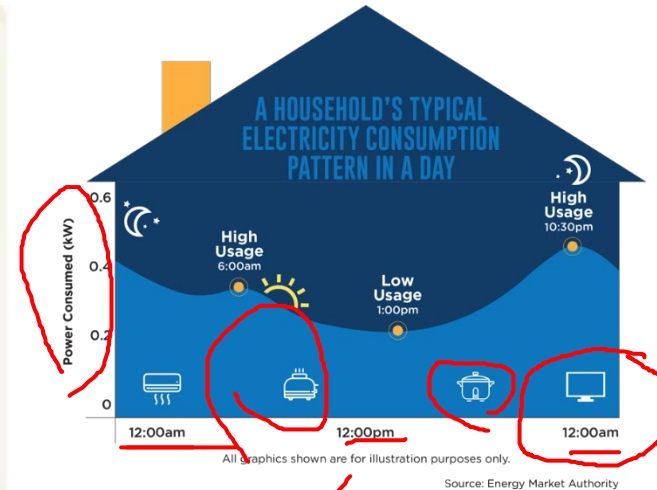
Example smart metering systems:



(a) Google PowerMeter

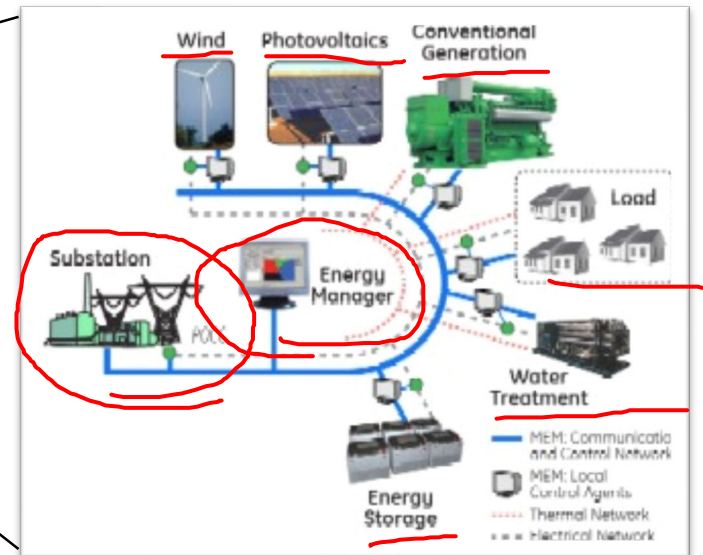
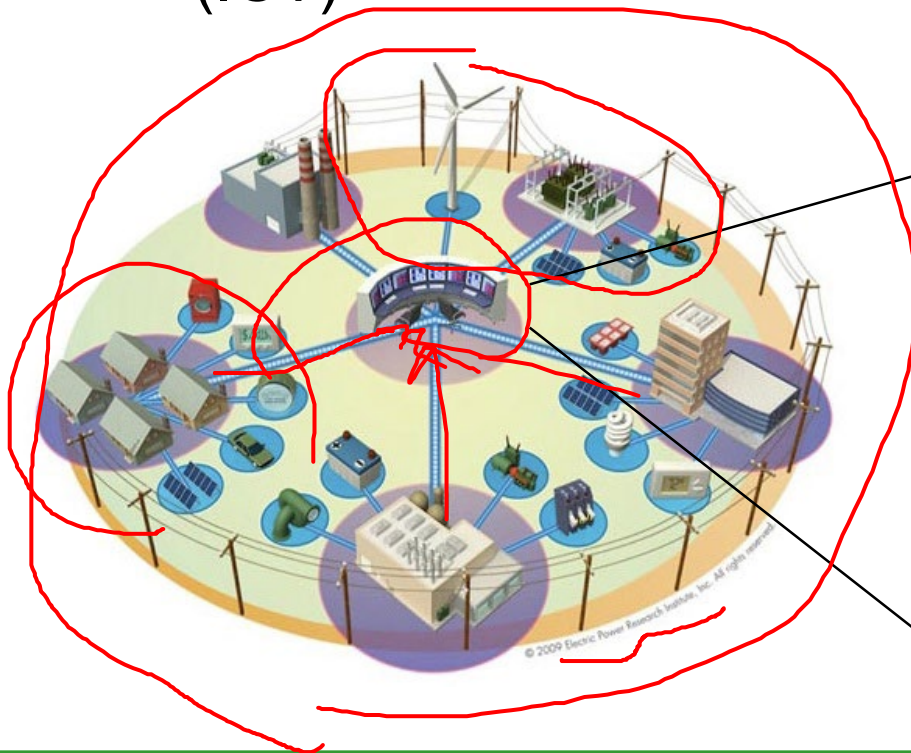


(b) Microsoft Hohm



Monitoring and Controlling

- Supervisory Control And Data Acquisition (SCADA)
- Energy management system (EMS)
- Information and Communications Technology (ICT)



Telecontrol

- Different protocols for different operations
 - Proprietary protocols (more than 100)
 - Standards
 - Supervisory control and data acquisition (SCADA)
 - Modbus (<https://modbus.org/>) 
 - Distributed Network Protocol 3 (DNP3) ←
 - A set of communications protocols used between components in process automation systems
 - IEC61850
 - Data Modeling, Reporting, Fast Transfer of events, Sampled Data Transfer, Commands, Data Storage



General Protocols

- MODBUS -Primitive without security and not very extensible
- DNP3 –Advanced SCADA protocol
- DNP1 and DNP2 are ~~proprietary protocols~~
- IEC 61850 the most used protocol for new implementations
- Inter-Control Center Communications Protocol (ICCP)



Outline

- Introduction to Smart Grid
- **Smart Grid Communication Architecture**
- Use Cases



NIST Interoperability in Smart Grid

- Physical interoperability and conventional interoperability
- Interoperability and cybersecurity
- Interoperability and emerging economic opportunities
- Standards and interoperability
- Device heterogeneity and interoperability

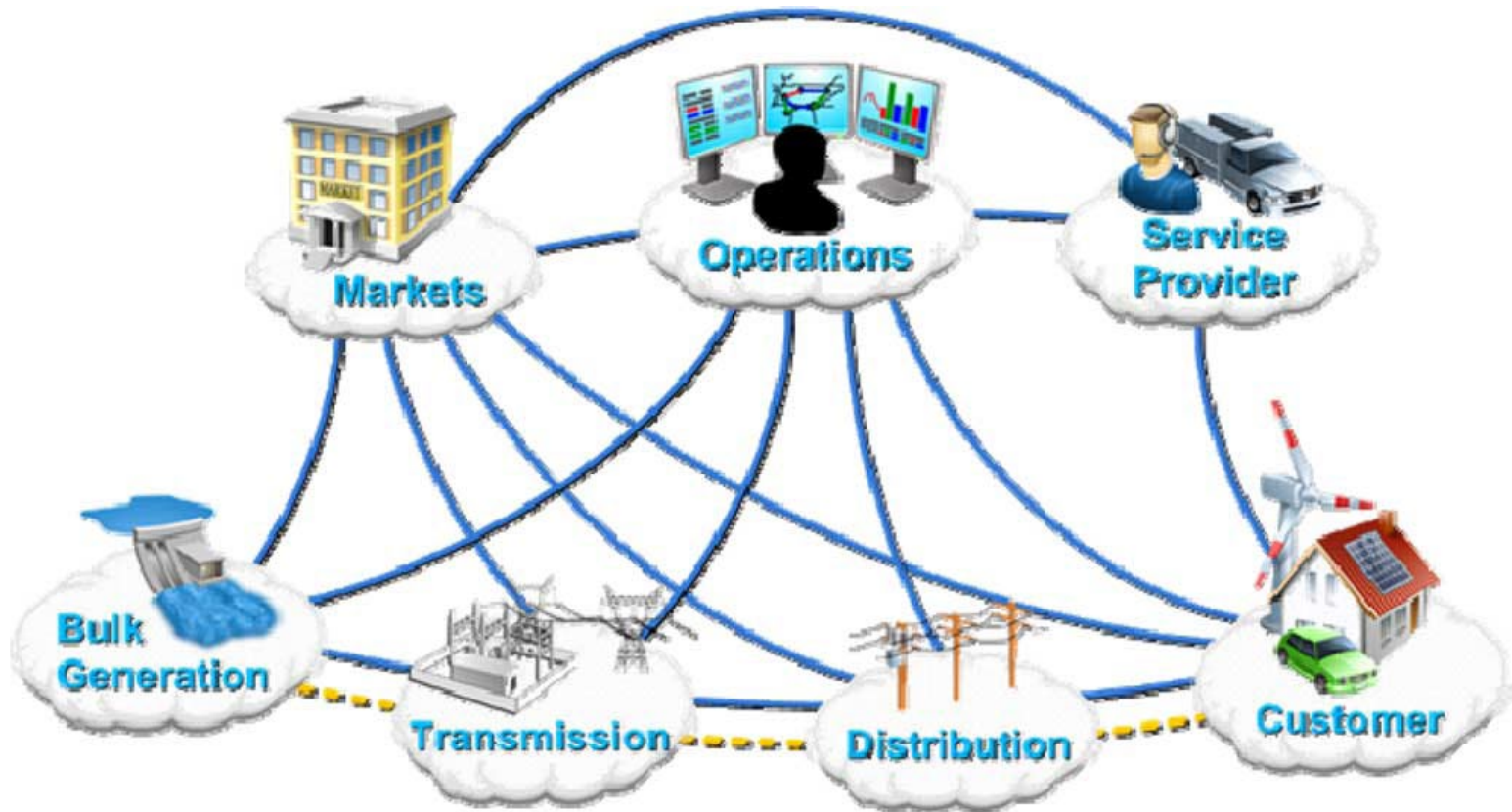
National Institute of Standards and Technology

<https://www.nist.gov/system/files/documents/2019/06/06/presentations-day1.pdf>

https://www.nist.gov/system/files/documents/public_affairs/releases/smartgrid_interoperability_final.pdf



NIST Conceptual Reference Model for Interoperability



National Institute of Standards and Technology

<https://www.nist.gov/system/files/documents/2019/06/06/presentations-day1.pdf>

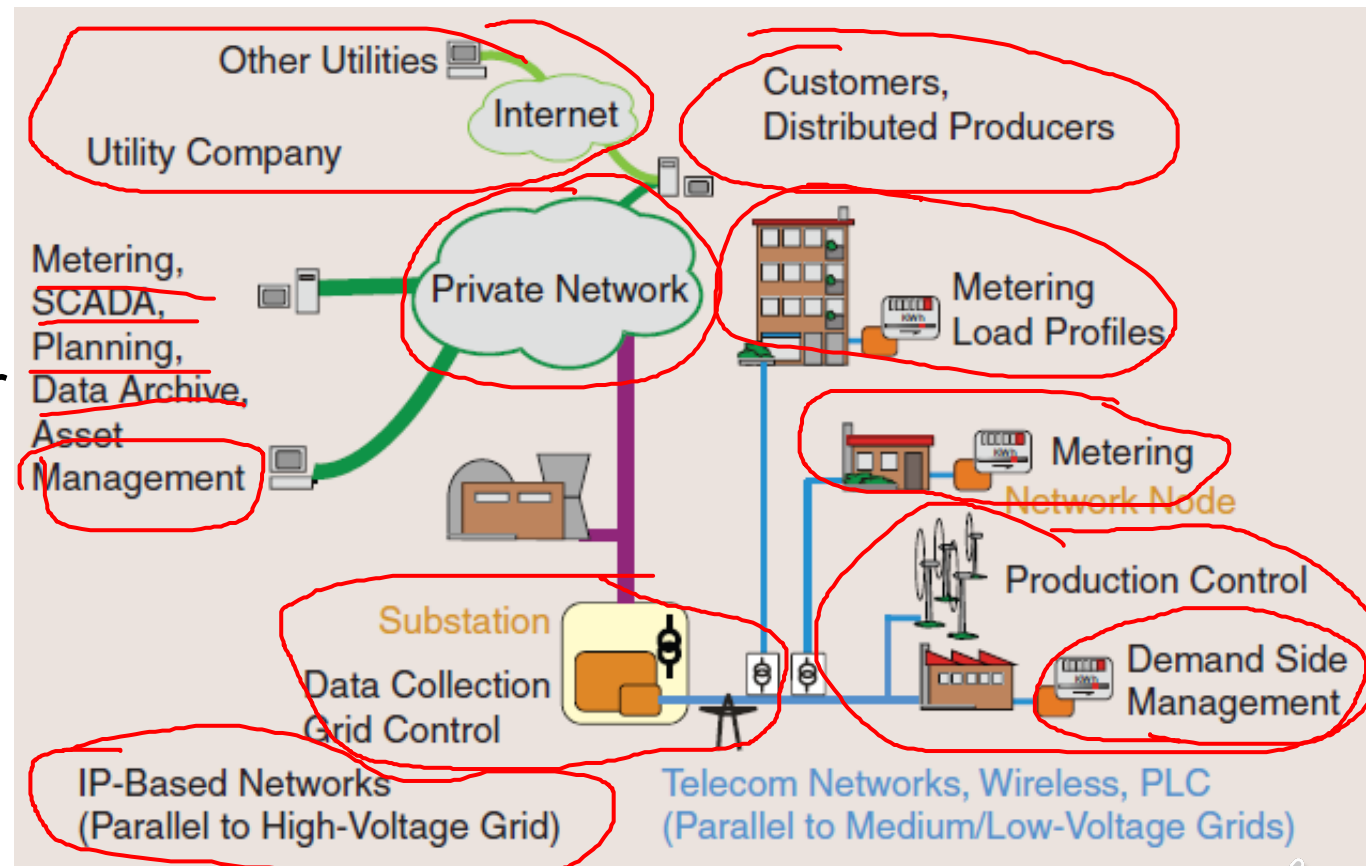
https://www.nist.gov/system/files/documents/public_affairs/releases/smartgrid_interoperability_final.pdf

Internet of Things (IoT) Case Study



Communication Media

- Urge for new FCC allocation for smart grids
- PLC –Power line carriers
- Ethernet
- WLAN
- ZigBee
- Bluetooth
- Optical fiber
- Microwave
- Etc.



Data Communications in Smart Grid

- Broadcast data (Demand Response, price signals, emergency events, etc.)
 - Low volume, infrequent
 - Can use currently available communication infrastructure (cellular, broadband, WiFi, Pager) with standard internet security measures
- Real-time Consumption Data (high volume, frequent)
 - Useful primarily for real-time control & usage information to consumer
 - We favor meter → premises where displays & controllers can locally act upon this data along with pricing information
 - Minimizes risk (privacy & network stability) and maximizes benefit from real time info.
- Raw Billing Data (reading when price changes)
 - Utility operations
- Aggregate Data
 - Comparison over time & among neighbors, best practices, consumption pattern recognition, suggest corrective actions, etc.
 - Utility or third party cloud-based applications operating on anonymous summary data
 - Little risk for privacy or network stability in case of breach of security
 - Can use standard internet communication with standard security measures
- T&D: relatively few points (substations); mission-critical, but already connected
Transmission and Distribution (T&D)

Optimal Network(s)

Broadband
Cellular
WiFi, etc.

Direct meter
to HAN

Home Area
Network (HAN)

AMI

Advanced Metering
Infrastructure (AMI)

Internet portal

Existing
connectivity



Smart Grid Communications Requirements

- Latency ✓
- Bandwidth ✓
- Interoperability ✓
- Scalability ✓
- Security ✓
- Standardization ✓

Latency

- The real-time operational data communications in smart grid include online sensor/meter reading and power system control signals.
- The communication is characterized by the fact that most of interactions must take place in real time, with hard time bound.
- The communication requirements define the design of the technical solutions.
- For real-time sensing/metering purposes, reading messages should be transmitted within a very short time frame.
 - For instance, the maximum allowed time is in the range of 12-20 ms, depending on the type of protection scheme which originates from the fact that the disconnection of fault current should be within approximately 100 ms.
- Power System Control signals mainly include supervisory control of the power process on secondary or higher levels. These systems are of the kind SCADA/EMS.
 - Measured values must not be older than 15 seconds, when arriving at the control center. Breaking information shall arrive no later than 2 seconds after the emergency event has occurred



Bandwidth

- As more and more interconnected intelligent elements are added to the electricity network with the evolution of the smart grid, the communication infrastructure should be able to transport more and more messages simultaneously without severe effect on latency.
- The network bandwidth must increase faster than the demand of these interconnected intelligent elements in the network.
- **An Example:** (A. Aggarwal, S. Kunta, P. K. Verma, "A proposed communications infrastructure for the smart grid," in Innovative Smart Grid Technologies (ISGT), 2010, pp. 1-5.)
 - Model the communication bandwidth requirements for a moderate size electricity distribution system. In this model, a distribution substation is connected to 10,000 feeders and each feeder connects to 10 customers.
 - Assuming that every electric meter generates a message every second to the distribution substation, the total is 100,000 messages per second. The feeders themselves will generate messages to each other and to the distribution substation.
 - The authors in this paper modeled the messages in the smart grid arriving at servers located at the control center as M/M/1 traffic. Then, the transmission line bandwidth is evaluated over 100 Mbps through the M/M/1 queuing model. It can be observed that this situation results in a very poor bandwidth utilization of the transmission facilities as well.
 - Unfortunately, a higher level of utilization will not permit meeting the assumed latency constraint.



Interoperability

- The ability of two or more networks, systems, devices, applications, or components to communicate & operate together effectively, securely, & without significant user intervention
 - Communication requires agreement on a physical interface & communication protocols
 - Exchanging meaningful & actionable information requires common definitions of terms & agreed upon responses
 - Consistent performance requires standards for the reliability, integrity, and security of communications
 - Interoperability may include:
 - “Plug and play”: connect them & they work together
 - Interchangeability: Ability to readily substitute components



Summary of Requirements

CURRENT FUNCTIONAL REQUIREMENTS						
Application	Security	Bandwidth	Reliability	Coverage	Latency	Back-up Power
Advanced Metering Infrastructure	High	14-100 kbps per node	99.0-99.99%	20-100 %	2000 ms	0-4 hours
AMI Network Management	High	56-100 kbps	99.00%	20-100%	1000-2000 ms	0-4 hours
Automated Feeder Switching	High	9.6-56 kbps	99.0-99.99%	20-100%	300-2000 ms	8-24 hours
Capacitor Bank Control	Medium	9.6-100 kbps	96.0-99.00%	20-90%	500-2000 ms	0 hours
Charging Plug-In Electric Vehicles	Medium	9.6-56 kbps	99.0-99.90%	20-100%	2000 ms - 5 min.	0 hours
Demand Response	High	56 kbps	99.00%	100%	2000 ms	0 hours
Direct Load Control	High	14-100 kbps per node	99.0-99.99%	20-100 %	2000 ms	0-4 hours
Distributed Generation	High	9.6-56 kbps	99.0-99.99%	90-100%	300-2000 ms	0-1 hour
Distribution Asset Management	High	56 kbps	99.00%	100%	2000 ms	0 hours
Emergency Response	Medium	45-250 kbps	99.99%	95%	500 ms	72 hours
Fault Current Indicator	Medium	9.6 kbps	99.00-99.999%	20-90%	500-2000 ms	0 hours
In-home Displays	High	9.6-56 kbps	99.0-99.99%	20-100%	300 - 2000 ms	0-1 hour
Meter Data Management	High	56 kbps	99.00%	100%	2000 ms	0 hours
Network Protection Monitoring	Medium - High	56-100 kbps	99.00-99.999%	100%	2000-5000 ms	0 hours
Outage Management	High	56 kbps	99.00%	100%	2000 ms	0 hours
Price Signaling	Medium	9.6-56 kbps	99.0-99.90%	20-100%	2000 ms - 5 min.	0 hours
Real-time Pricing	High	14-100 kbps per node	99.0-99.99%	20-100 %	2000 ms	0-4 hours
Remote Connect/Disconnect	High	56-100 kbps	99.00%	20-100 %	2000-5000 ms	0 hours
Routine Dispatch	Medium	9.6-64 kbps	99.99%	95%	500 ms	72 hours
Transformer Monitoring	Medium	56 kbps	99.00-99.999%	20-90%	500-2000 ms	0 hours
Voltage and Current Monitoring	Medium	56-100 kbps	99.00-99.999%	100%	2000-5000 ms	0 hours
Workforce Automation	Medium	256-300 kbps	99.90%	90%	500 ms	8 hours

National Broadband Plan: RFI Communications Requirements

Comments of Utilities Telecom Council, July 12, 2010

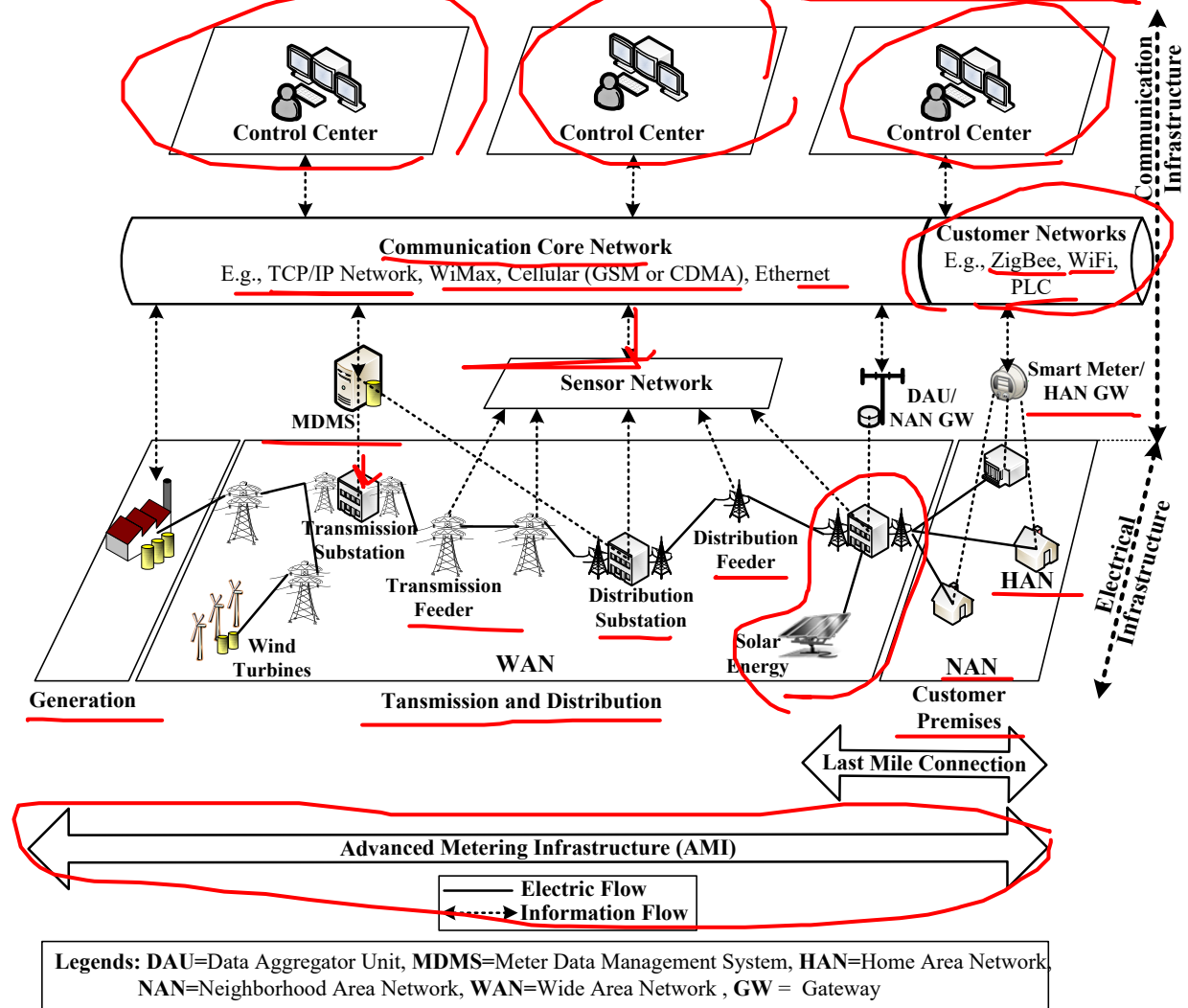


Communication Architectures

- Communication Architecture and Model for Distribution Network
- Home-Area Networks (HANs)
- Neighborhood-Area Networks (NANs)
- Wide-Area Networks (WANs)
- Sensor and Actuator Networks (SANETs)

Communication Architectures

Communication Architecture and Model for Distribution Network



Communication Architectures

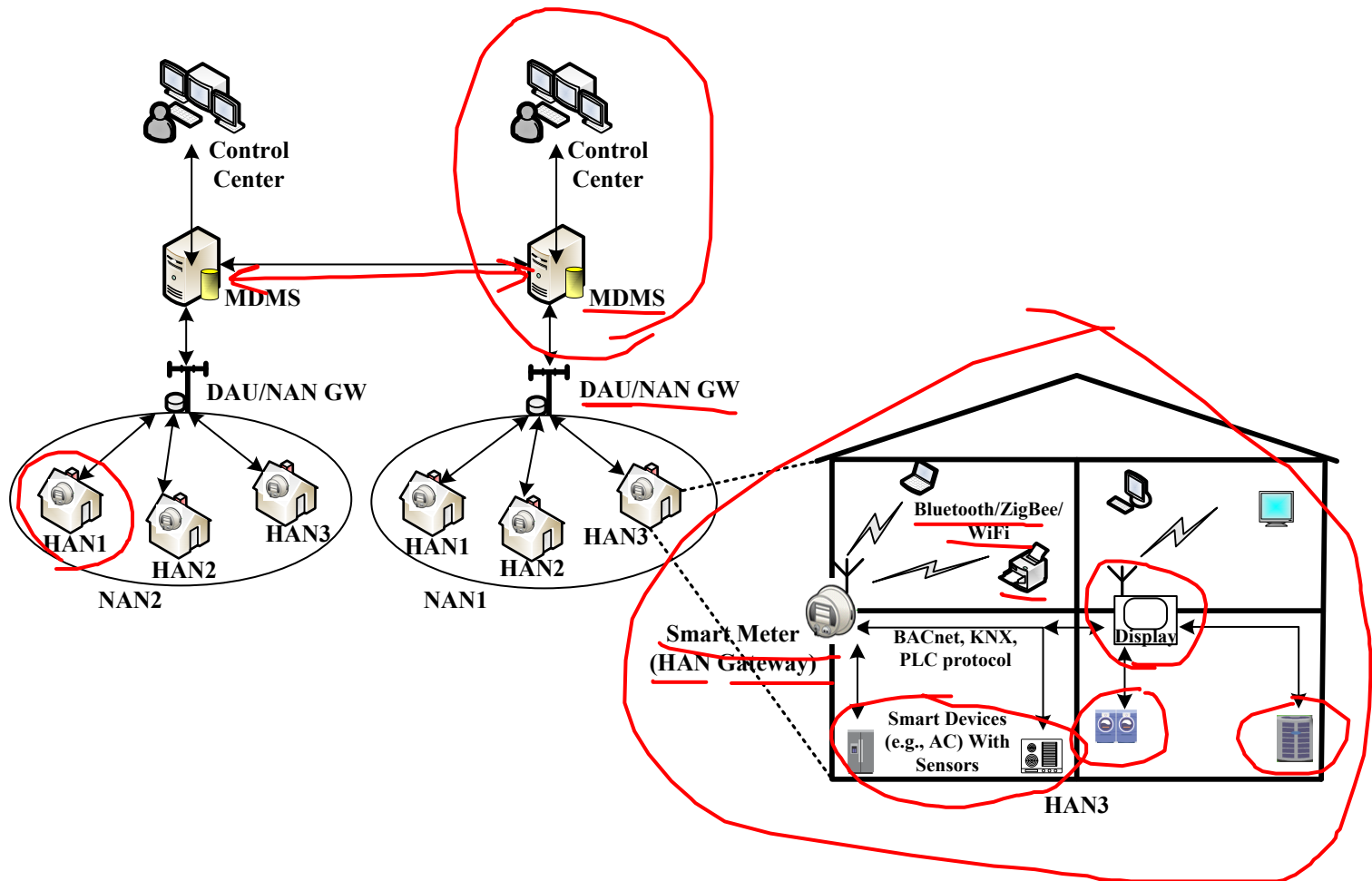
Communication Architecture and Model for Distribution Network

- Smart grid follows the same electrical architecture
- Electricity is delivered from the generation to consumers through transmission and distribution substations
- Transmission substation delivers electricity from power generation plant over a high voltage transmission line (over 230kV) to the distribution substation
- Distribution substation converts the electric power to medium voltage level
- Distribution feeder then converts the medium voltage to lower level for distributing to the consumer's end



Communication Architectures

Customer Premise and Customer Network



Communication Architectures

Customer Premise and Customer Network

- Data aggregator unit (DAU) also referred to as NAN GW acts as a data sink to collect and relay the information from the consumer side to meter data management system (MDMS)
- MDMS will provide storage, management, and processing of meter data for proper usage by other power system applications and services



Communication Architectures

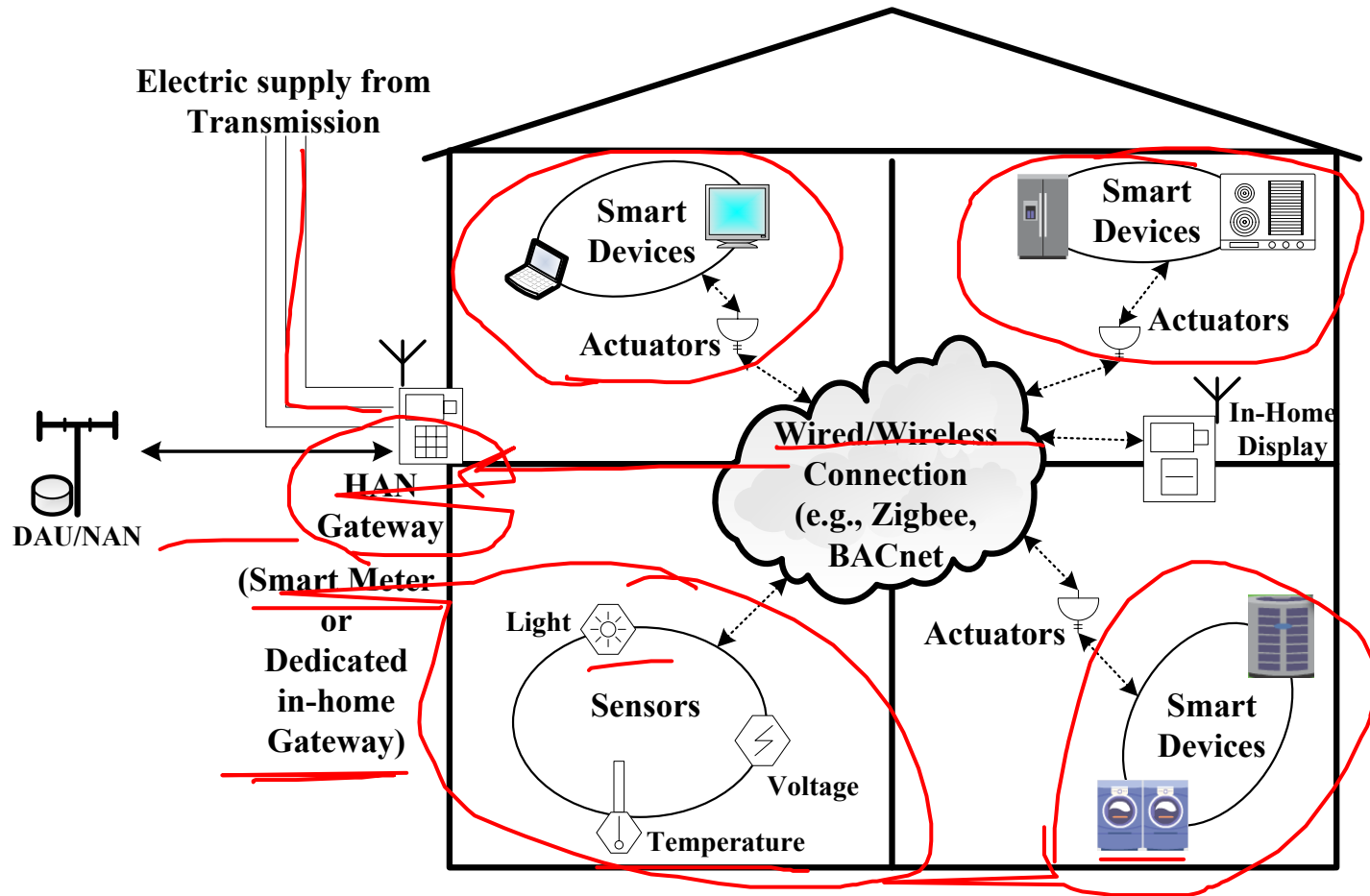
Home-Area Networks (HANs)

- HAN (sometimes referred to as Premise Area Network (PAN) or a Building Area Network (BAN)) is the smallest subsystem in the hierarchical chain of smart grid
- HAN provides a dedicated demand side management (DSM), including energy efficiency management, and demand response by proactive involvement of power users and consumers
- HAN consists of smart meter, smart devices with sensors and actuators, and in-home display for energy management system (EMS)
 - EMS will provide means of reducing energy consumption by monitoring and controlling different electrical appliances



Home-Area Networks (HANs)

General Structure



Home-Area Networks (HANs)

Enabling Communications Technologies

- Short Range Wireless Technologies
 - Wi-Fi, Bluetooth, ZigBee, Z-Wave
- Z-Wave:
 - Proprietary wireless standard designed for home control automation, specifically to remote control applications in residential homes
 - Z-Wave was originally developed by Zensys A/S and is being marketed by Z-Wave Alliance
 - Z-Wave wireless protocol provides reliable and low-latency communication of small data packets within HANs
 - Z-Wave also uses a mesh networking approach with source routing



Home-Area Networks (HANs)

Enabling Communications Technologies: Wireless Technologies

Key Criteria	WiFi	Bluetooth	ZigBee	Z-Wave
Feature	- Designed for providing wireless connection for accessing Internet and is direct replacement to traditional Ethernet network	- Designed for consumer electronics to provide short-range wireless communication to connect a wide range of devices easily and quickly	- Designed specifically for industrial and home automation for connecting sensors, monitors and control devices	- Designed for home automation, specifically to remote control applications in residential home such as light, entertainment systems, etc
Frequency Band	- 2.4/5 GHz	- 2.4 GHz	- 2.4 GHz, 915MHz and 868MHz	- 900 MHz
Standards	- International Standard (IEEE 802.11 a/b/g/n) - Open Standard	- International Standard (IEEE 802.15.1) - Open Standard	- International Standard (IEEE 802.15.4) - Open Standard	- Proprietary Standard (Z-Wave Alliance and Zensys) - Closed standard
Speed	- 54 Mbps (802.11. b/g) - 150 Mbps (802.11 n)	- 2.1 Mbps (V 2.0) - 20 Mbps (V 3.0, recently released)	- 250 Kbps	- 9600 bits/s
Range	- 70m (indoor) to 250m (outdoor)	- 10m	- 70m (indoor) to 400m (outdoor)	- 30m (indoor) to 100m (outdoor)
Power Consumption	- High	- Lower than WiFi	- Lower than WiFi and Bluetooth	- Almost same as ZigBee
Maximum Nodes	- 2007	- 8	- > 64000	- 232

Home-Area Networks (HANs)

Enabling Communications Technologies: Wireless

Key Criteria	WiFi	Bluetooth	ZigBee	Z-Wave
Security	- WEP (Wired Equivalent privacy) - WPA (Wi-Fi Protected access) - WPA2	- E0 stream cipher - More Secure than WiFi	- 128 AES (Advanced Encryption Standard) keys	- 3 DES(Triple Data Encryption Standard)
Strength	- Easy to deploy, equipment costs dropping rapidly - Supports mesh topology	- Most popular protocol for transferring data and wireless alternative to RS-232 data cables - Supports ring topology	- Low power requirements and implementation costs - Particularly designed for use in industrial and home automation or security applications - Scalable and flexible - Supports mesh topology	- Low power, low latency, and low cost - Less interference due to use of sub-GHz frequency - Higher propagation range 2.5 times the 2.4 GHz signal - Supports mesh topology
Concern	- High power consumption - Higher data latency - Additional security layer should be implied to use WiFi within HAN	- Lack of proper installation in consumer portal context such as fire alarm, security sensors, etc - Does not support mesh networking	- Limited range and low data rates - Interference due to overlapping with WiFi standard	- Low data rates - Requires to add devices into network manually - Slightly higher installation cost than ZigBee - Offers less flexibility due to close nature

Home-Area Networks (HANs)

Enabling Communications Technologies: Wired Technologies

Key Criteria	X10	HomePlug GP	BACnet	KNX
Feature	- Simple and popular protocol designed for providing simple automation functionality such as on and off.	- Designed specifically for smart grid to provide lower power consumption	- A data communication protocol that attempts to unify all the proprietary communication protocols into single communication language	- Global standard protocol designed basically for home automation and control
Wireless Support	- Yes - 310 MHz U.S. 433 MHz European	- Yes - Recent ZigBee/HomePlug initiatives	- Yes	- Yes - KNX RF (868.3 MHz)
Standards	- De facto Standard - Open Standard	- International Standard (IEEE 1901)	- ANSI/ASHRAE 135-2008 - ISO 16484-5 - Open Standard	- CENELEC EN 50090 and CEN EN 13321-1 - ISO/IEC 14543-3 - GB/Z 20965, ANSI/ASHRAE 135
Speed	- 20 bits/s	- 4 to 10 Mbps	- Depends on choice of LAN technology used	- wired 9.6 kbps - wireless 16.4 kbps
Maximum Nodes	- 256	- 253 (theoretically) - 10 (Practically)	- No limit	- 57600 network nodes for wired connection

Home-Area Networks (HANs)

Enabling Communications Technologies: Wired

X10 Technologies

HomePlug GP

BACnet

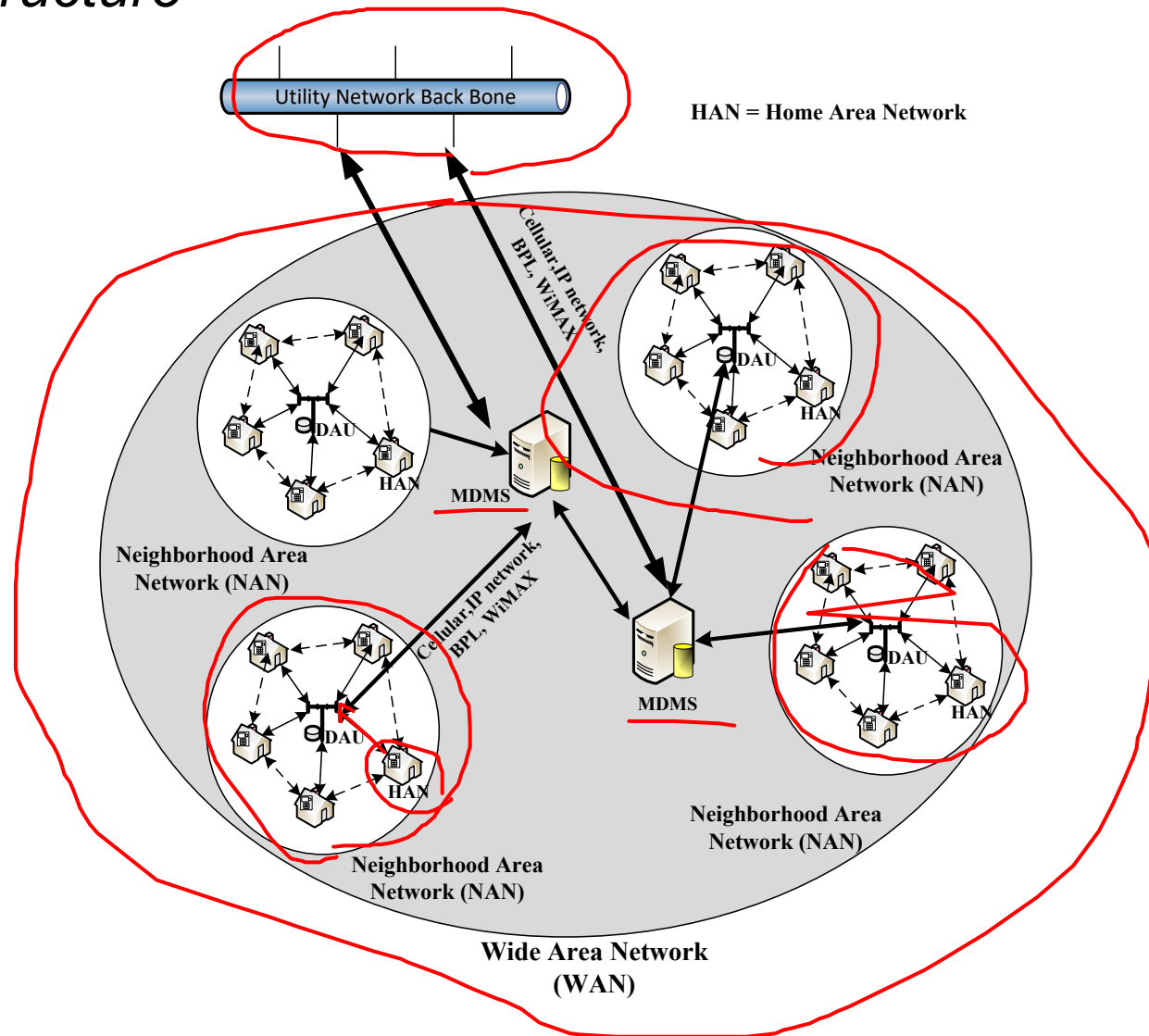
KNX ✓

Key Criteria	<u>X10</u> Technologies	<u>HomePlug GP</u>	<u>BACnet</u>	KNX ✓
<u>Security</u>	- lack of encryption ✓	- AES pro 128 Security (128-bit triple AES encryption and a time lock) ✓	- Assume that all devices are sitting behind a firewall ✓	- EIBsec ✓
<u>Strength</u>	- Commonly used with variety of equipment available in the market - <u>No installation cost as uses power line</u>	- Ubiquitous reach throughout the home environment ✓ - Interoperability with consumer home networking - Low-cost and low-power network interfaces ✓ - Cross-compatibility between wired and wireless Smart Grid applications	- Well established as an enabler for commercial building automation technologies ✓ - Already has the needed functionality for energy management and load control ✓ - Independent of current LAN or WAN technologies ✓ - Scalable ✓	- Interoperable with other KNX products ✓ - Hardware/software independent ✓ - Well established promoter which provide any application for home control ✓ - Compatible with any buildings ✓
Concern	- Extreme low data rate ✓ - Lack of standard and security ✓ - Limited functionality ✓ - Prone to interference from neighbors using the same X10 device addresses ✓	- Limited connection (10) when transferring data simultaneously ✓ - Susceptible to power line interference and old wiring in home. ✓	- client-server system might create bottleneck when fully deployed to consumer premise ✓ - Security concerns ✓ - Object model is limited to low-level types ✓	- Low data rates ✓



Neighborhood-Area Networks (NANs)

General Structure



Neighborhood-Area Networks (NANs)

Neighborhood-Area Networks (NANs)

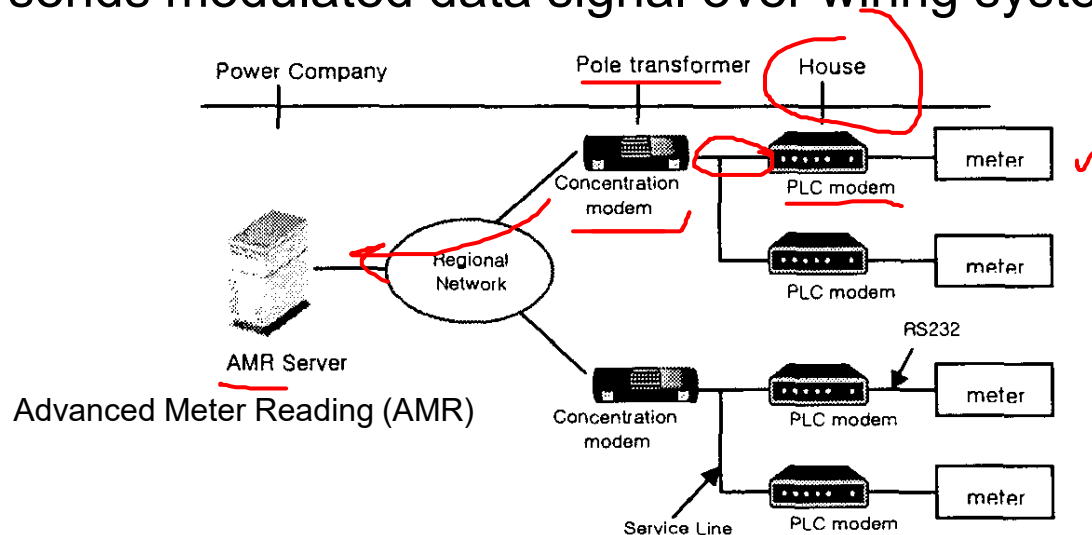
- NAN connects multiple HANs together
- Wired Technologies
 - Power Line Communication (PLC):
 - Ultra narrow band (UNB) operates in 0.3-3 kHz bands
 - Narrow band (NB) PLC operates in 3-500 KHz bands
 - Broadband (BB) PLC or BPL operates in 1.8-250 MHz bands
 - Internet Protocol (IP)-Based Networks
 - Internet Based Virtual Private Networks (Internet VPN)
 - Internet VPN technology can provide reliable, secure, and robust alternative to ensure security and QoS requirement
- Wireless Technologies
 - 3G and LTE cellular networks



Neighborhood-Area Networks (NANs)

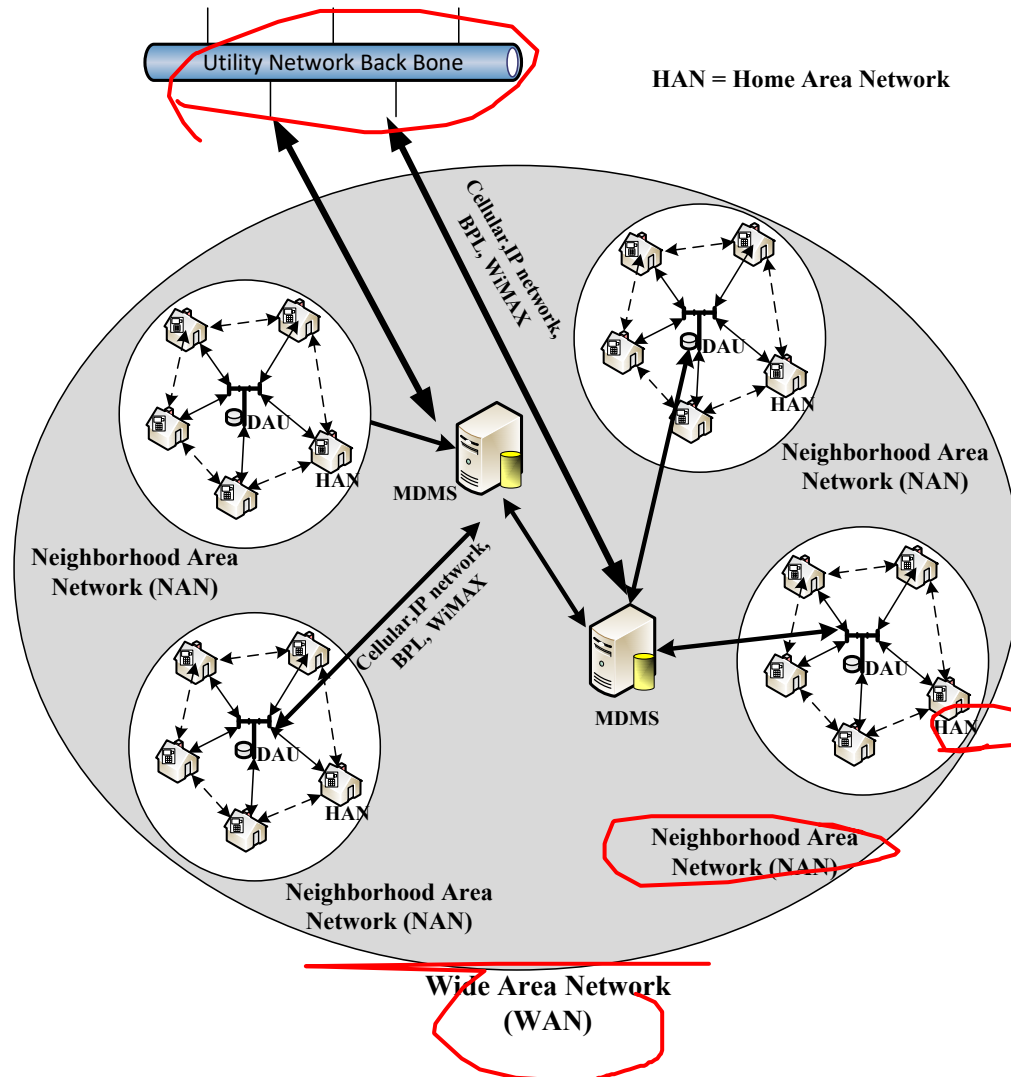
Neighborhood-Area Networks (NANs)

- Power Line Communication (PLC)
- PLC uses AC electrical power transmission wires or electric power distribution to consumers as medium for data transfer
- PLC can be used for home automation and Internet access applications, i.e. broadband over power lines (BPL)
- PLC sends modulated data signal over wiring system



Wide-Area Networks (WANs)

General Structure



Wide-Area Networks (WANs)

Core Communication Network and Last Mile Connectivity

Enabling Technologies	Scope	Strength	Concern
1. Power Line Communication	- Communication Core Network and Last Mile Connection	- Complete control over the communication path with extensive coverage that is solely controlled by the utility industry - Provides low cost solution to overlay the communication network over already available power lines - Provides direct route between controllers and other subsystem to ensure low latency - Mature technology with many variants available commercially	- The power line are connected to various equipments such as motor, power supplies, which can act as noise sources that eventually degrades the performance of PLC - The load impedance fluctuation, and electromagnetic interference causes signal attenuation and distortion, which can result to failure of communication link - Lack of standard status and government regulation due to industry fragmentation result in high interference from other PLC technology deployed at close range - Cost of PLC modem are still high - Coexistence issue from many commercial technologies
2. Internet Protocol (IP)-Based Networks	- Communication Core Network and Last Mile Connection	- IP-based networks have rich convergence capabilities which can help to connect the overall systems and subsystems in smart grid - Can provide QoS and reliable connection using technologies such as DiffServ and MPLS - Security can be enhanced using technologies (IPSec)	- In case of master/slave configuration, transmitting IP packets from slave is not possible, which might increase the data latency for those applications which requires fast response as in case of smart grid - Unless private IP-based network (e.g., Internet VPN) is used, security remains crucial issue



Wide-Area Networks (WANs)

Core Communication Network and Last Mile Connectivity

Enabling Technologies	Scope	Strength	Concern
3. Wireless Communication	- <u>Communication Core Network and</u> Last Mile Connection	- <u>Huge coverage area, potential for low cost</u> - <u>Packet-Switched Cellular Data</u> has lower cost and <u>much higher data rates</u> - <u>WiMAX</u> can support mesh networks for higher reliability	- <u>Utilities have to depend on these technologies</u> without any control over them - <u>Packet switch technologies</u> are not available in all <u>deployed cellular structures</u> - Requires connection to network before transmitting the data and might be problem in case of <u>outage and</u> emergency

Communication Architectures

Standard Activities (1)

Standards	Application	Strength	Concern
ANSI C12 Suite ANSI C12.19/IEEE 1377 ANSI C12.22	- Defines utility industry end device data tables for representing the data produced by revenue meters. - Standard protocol for network communication	- Defines format of data for meter - Provides transport independent application level protocol for data exchange with low overheads between nodes. - Supports transport of C12.9 table data - Provide authentication & encrypting the C12.9 data.	- Does not specify protocol to transport it -Lacks full interoperability due to specialized local profile - Requires complexity in implementation in clients
ANSI/ASHRAE 135/ISO 16484-5 BACnet	- Defines information model & messages as objects for providing common language for different proprietary protocols	- Open, mature standard with interoperability testing developed and maintained by SDOs - Serves as customer side communication protocol with relevancy in price, DR/DER & energy usage	- Object model might be limited to low level protocols - Requires structural view & specific profile to address consumer portals
ANSI /EIA/CEA 709 & CEA 8521 Protocol Suite LONworks ANSI/CEA 709.1-B ANSI/CEA 709.2 ANSI/CEA 709.3 ANSI/CEA 709.4	- General purpose LAN protocol for providing communication over with home & building automation - The Control Network - Power Line Carrier Physical Layer - Twisted Pair Physical Layer - Fiber Optic Physical Layer	- Widely used matured protocol - Specify as one of the data link & physical layer option for BACnet	- de facto standard controlled by Echelon with limited support in power industry - Lack of complex object model to support function



Communication Architectures

Standard Activities (2)

Standards	Application ✓	Strength ✓	Concern
<u>ZigBee/ HomePlug Smart Energy Profile</u>	Strategic alliance of ZigBee & HomePlug to provide communication & information model in HAN	- Interoperable between two distinct HAN technology - Technology independent	
<u>IEC 62056 Device Language Message Specification (DLMS) & Companion Specification for Energy Metering (COSEM)</u>	- Standard representation of metering data used for accessing and exchanging structured data models	- Supports object modeling of application data as object identification system (OBIS) and the Open Systems Interconnection (OSI) model - Matured and internationally recognized standard - Supports variety of media such as PSTN, GSM network, PLC and recently ZigBee protocols	
<u>IEEE 1901</u>	- Broadband communications over Powerline medium access control (MAC) and physical layer (PHY) Protocols for HAN and also access application	- High speed (>100 Mbps) communication for devices using frequency below 100 MHz. - Uses inter-system protocol (ISP), which allow device to coexist with devices based on ITU-T G.hn standard - Initiate harmonization and coexistence of PLC with other technologies - Has backward compatibility with HomePlug standard	- Short range due to higher attenuation of the medium as a result of using broadcast channels above 80 MHz ✓



Communication Architectures

Standard Activities (3)

Standards	Application	Strength	Concern
ITU-T G.hn/G.9960 Home Networking Standard	- In-home networking over power lines, phone lines, and coaxial cables	- Designed especially for HAN - Use single fast Fourier transform (FFT) OFDM modulation and low-density parity-check code (LDPC) forward error correction (FEC) code	- Does not address PLC access application - Does not support HomePlug standard
ISO/IEC 15045, A residential gateway model for Home electronic system	- Defines specification for residential gateway (RG) that connects HAN to network domain outside the home basically last mile connection	- Defines functional requirement & architecture for RG - Defines security requirements for connecting to WANs	- Still under consideration by independent organization
ISO/IEC 15067-3, Model for an energy management system for Home electronic system	- Defines a model for energy management system that accommodates a range of load control strategies	- Specifies methods for demand response that may be implemented by an electric utility or by a third-party supplier of energy management services - Supports various smart appliances	

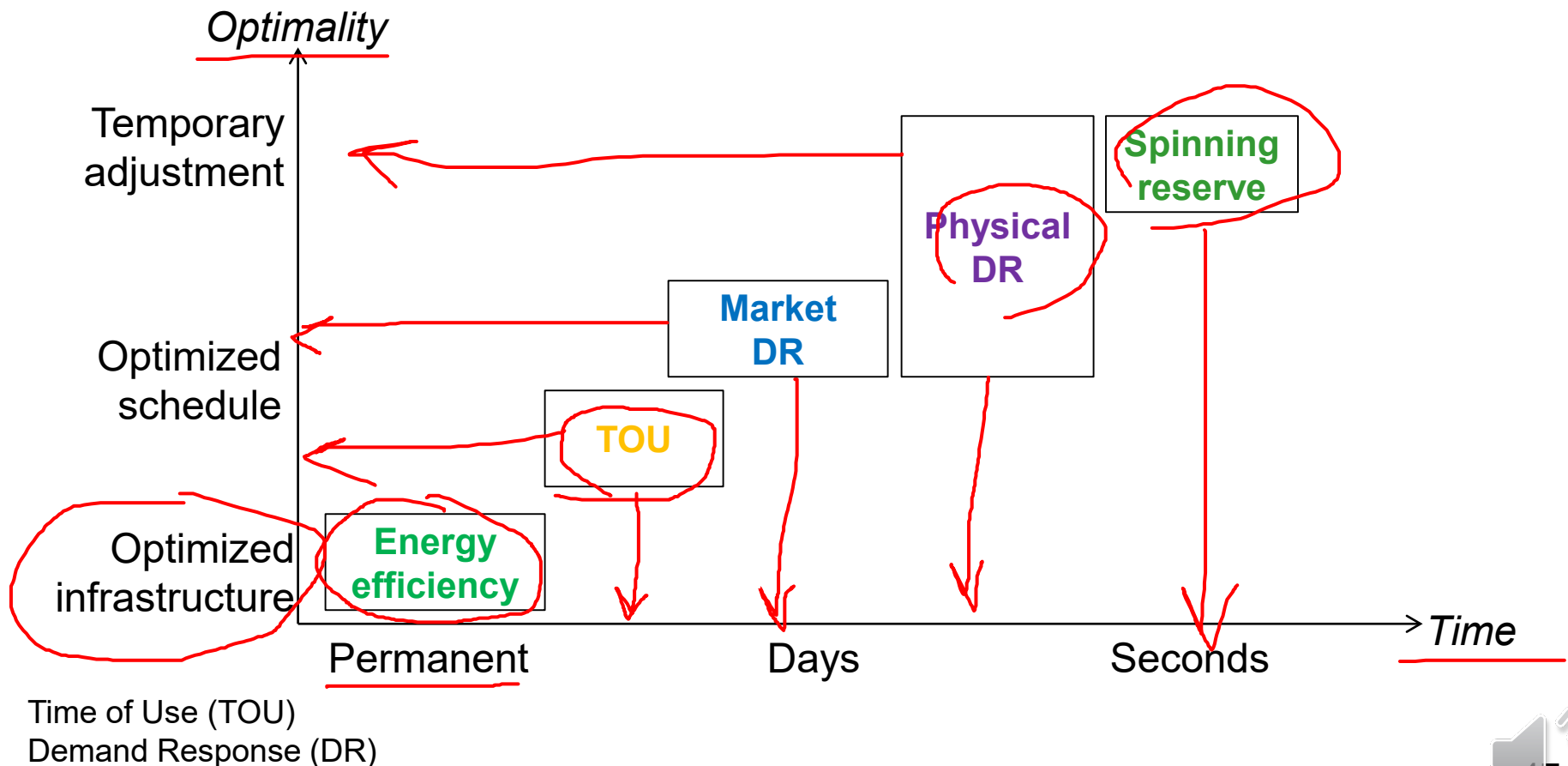
Outline

- Introduction to Smart Grid
- Smart Grid Communication Architecture
- **Use Cases**
 - **Demand Response Management (DRM)**
 - **Home Energy Management System (HEMS)**
 - **Advanced Metering Infrastructure (AMI)**



Demand Response Management (DRM)

- DRM is the programs implemented by utility companies to control the energy consumption at customer side



Demand Response Management (DRM)

- Energy efficiency focuses on users and behavioral changes to achieve more efficient energy usage
 - Users buy appliance with energy reduction feature



Energy		Washing machine
Manufacturer Model		
More efficient		
A		
B		
C		
D		
E		
F		
G		
Less efficient		
Energy consumption kWh/cycle (based on standard test results for 60°C cotton cycle) Actual energy consumption will depend on how the appliance is used		1.75
Washing performance A: higher G: lower		A B C D E F G
Spin drying performance A: higher G: lower Spin speed (rpm)		A B C D E F G 1400
Capacity (cotton) kg		5.0
Water consumption		5.5
Noise (dB(A) re 1 pW)		Washing 5.2 Spinning 7.6
Further information contained in product brochure		

Demand Response Management (DRM)

- Smart pricing or time of use (TOU)
 - Customers (re)arrange their energy consumption to minimize costs
- Market demand response
 - Direct load control (DLC): utility or grid operator control energy consumption of consumers
 - Interruptible/curtailable rates: customers has a contract with limited sheds feature from utility
 - Emergency demand response programs: customers voluntarily adjust energy consumption based on emergency signals (e.g., blackout)
 - Demand bidding programs: customers can bid for curtailing at attractive price



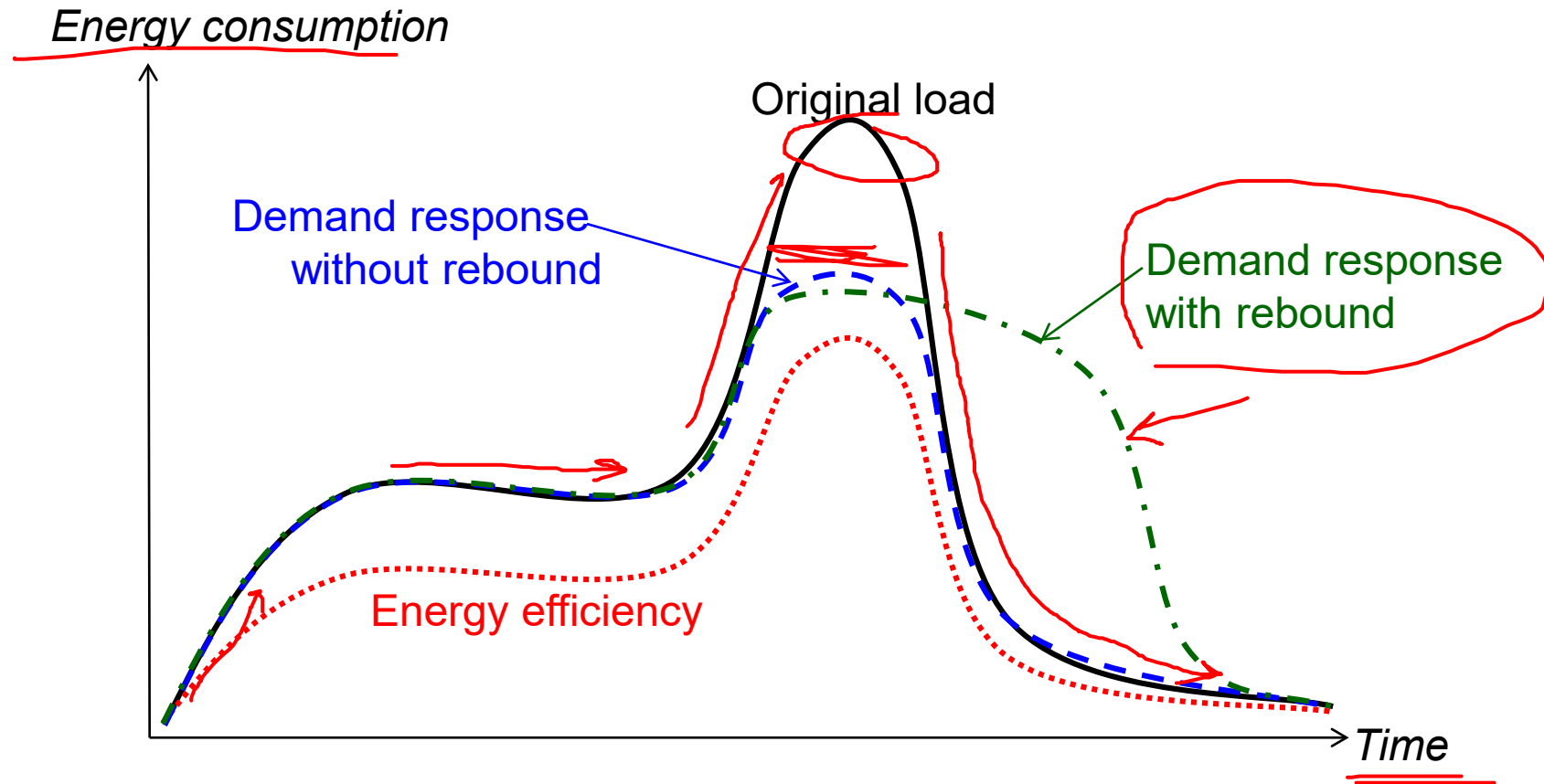
Demand Response Management (DRM)

- Physical demand response
 - Grid management and emergency signals (on the utility side)
 - Signal if the grid (power lines, transformers, and substations) are in a reduced performance due to maintenance or failure
- Spinning Reserves (SR)
 - Generators are online, synchronized to the grid, that can increase output immediately in response to a major outage and can reach full capacity



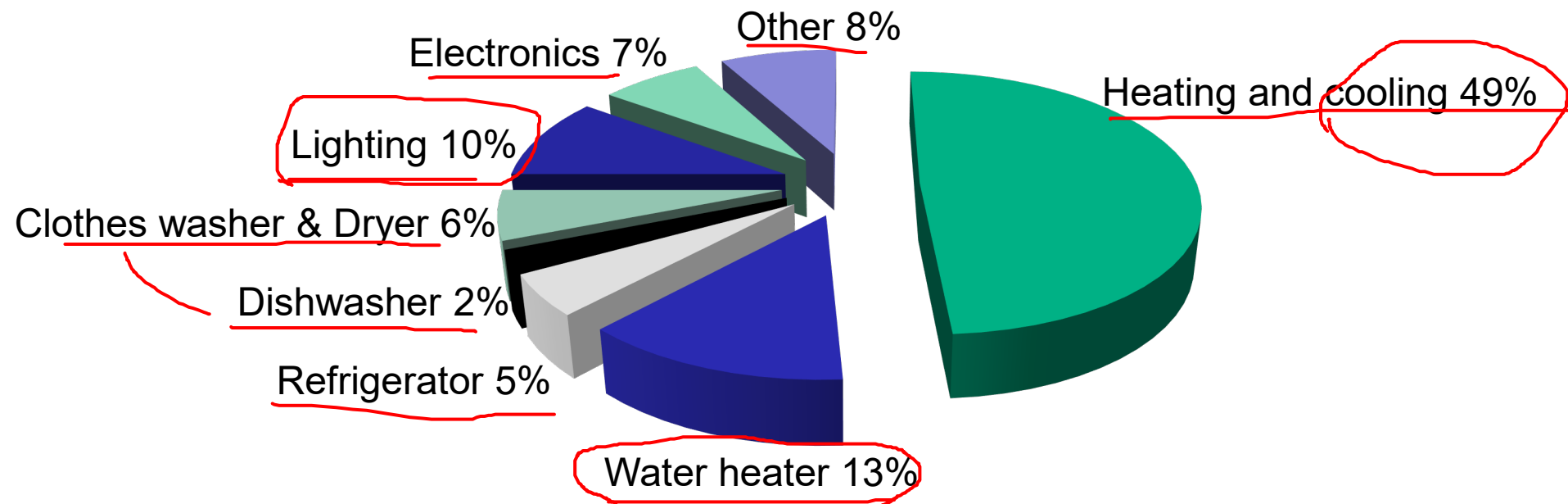
Demand Response Management (DRM)

- Energy efficiency vs. demand response



Home Energy Management System (HEMS)

- HEMS acts as the subset of energy management system (EMS) and together with smart meter provides a necessary interface to the HAN for better energy management



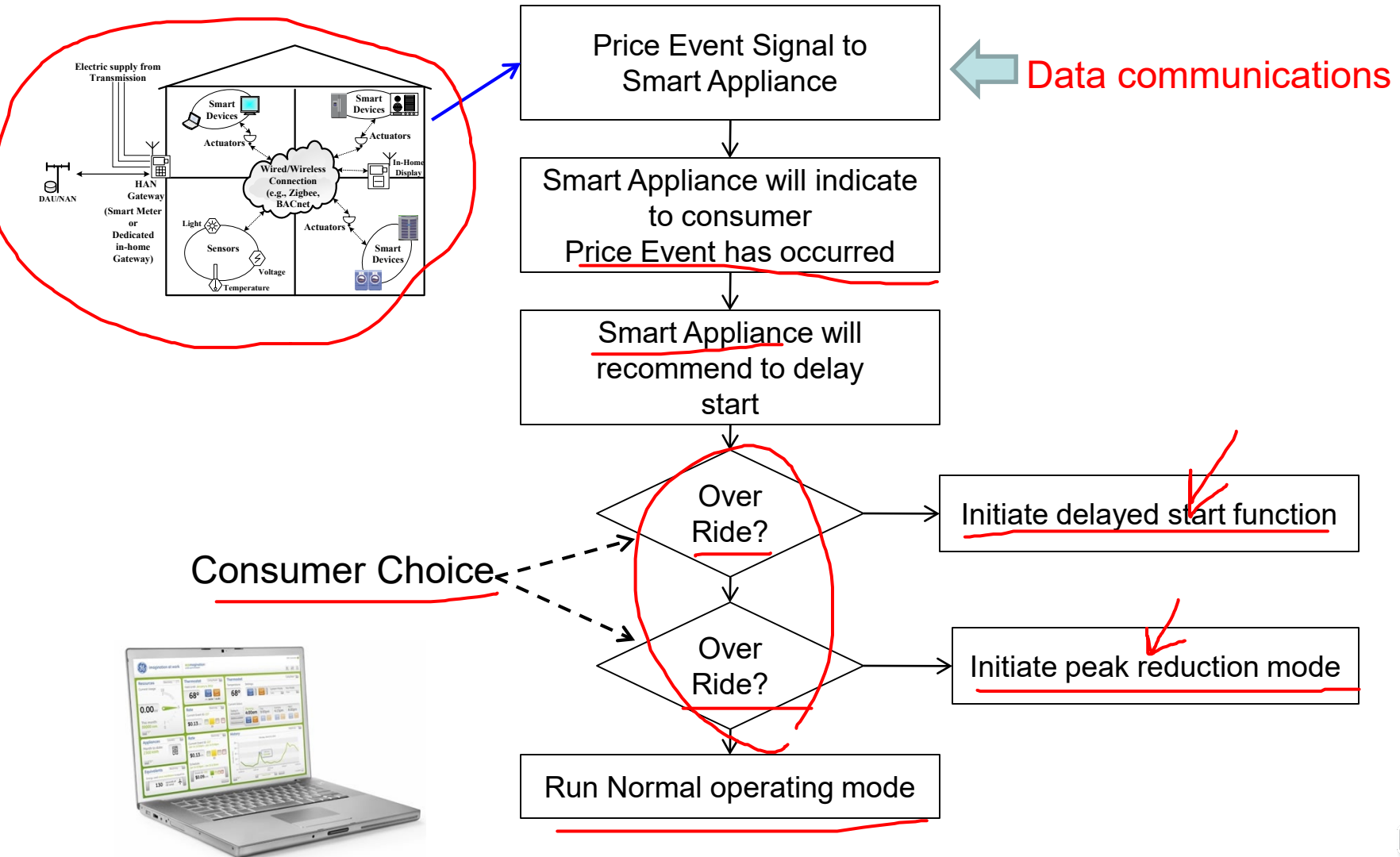
<http://www.energystar.gov/>

Home Energy Management System (HEMS)

- HEMS (or EMS) sets a certain user limit threshold based on the information about real-time price-responsive load management and consumption history (i.e., collected from smart meter) to control the energy usage of appliances
- HEMS is generally integrated into HAN to offer a channel for the consumers to interact with the electrical power grid
- HEMS may reside in the smart meter or in an independent gateway such as residential gateway and network adapters
- HAN connects many electrical appliances (routers, TV, AC, computers, etc.) which provide different services, e.g., wireless access, VoIP calls, ambient temperature control
- These services can be controlled by using different power control elements (PCEs) such as Wi-Fi, Ethernet switch, and power line communication.

Home Energy Management System (HEMS)

- Example: GE Demand Reduction Approach

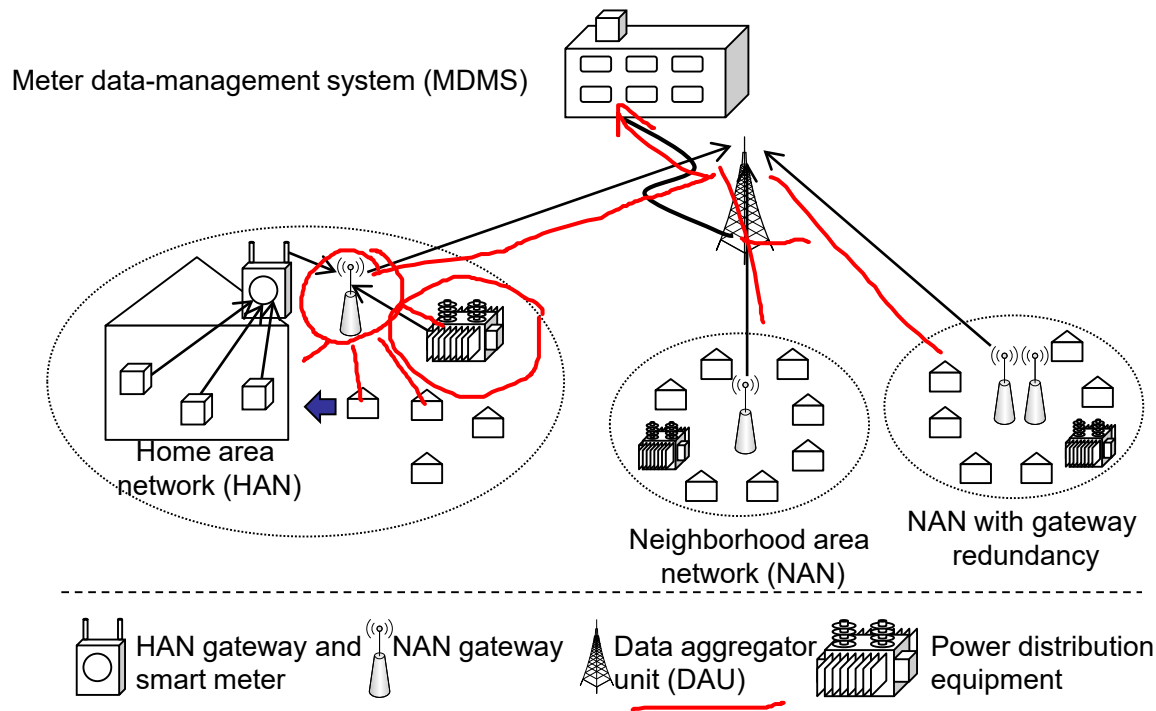


Advanced Metering Infrastructure

- AMI provides two-way communication for utility company to connect to consumers' smart meter
- AMI is used to collect meter reading more frequently, e.g. hourly basis
- Meter data is used for managing demand in a real-time fashion through demand side management (DSM)
- Data can be used for security purposes, e.g. detecting and preventing electricity theft

Advanced Metering Infrastructure

Reliability Analysis



Advanced Metering Infrastructure

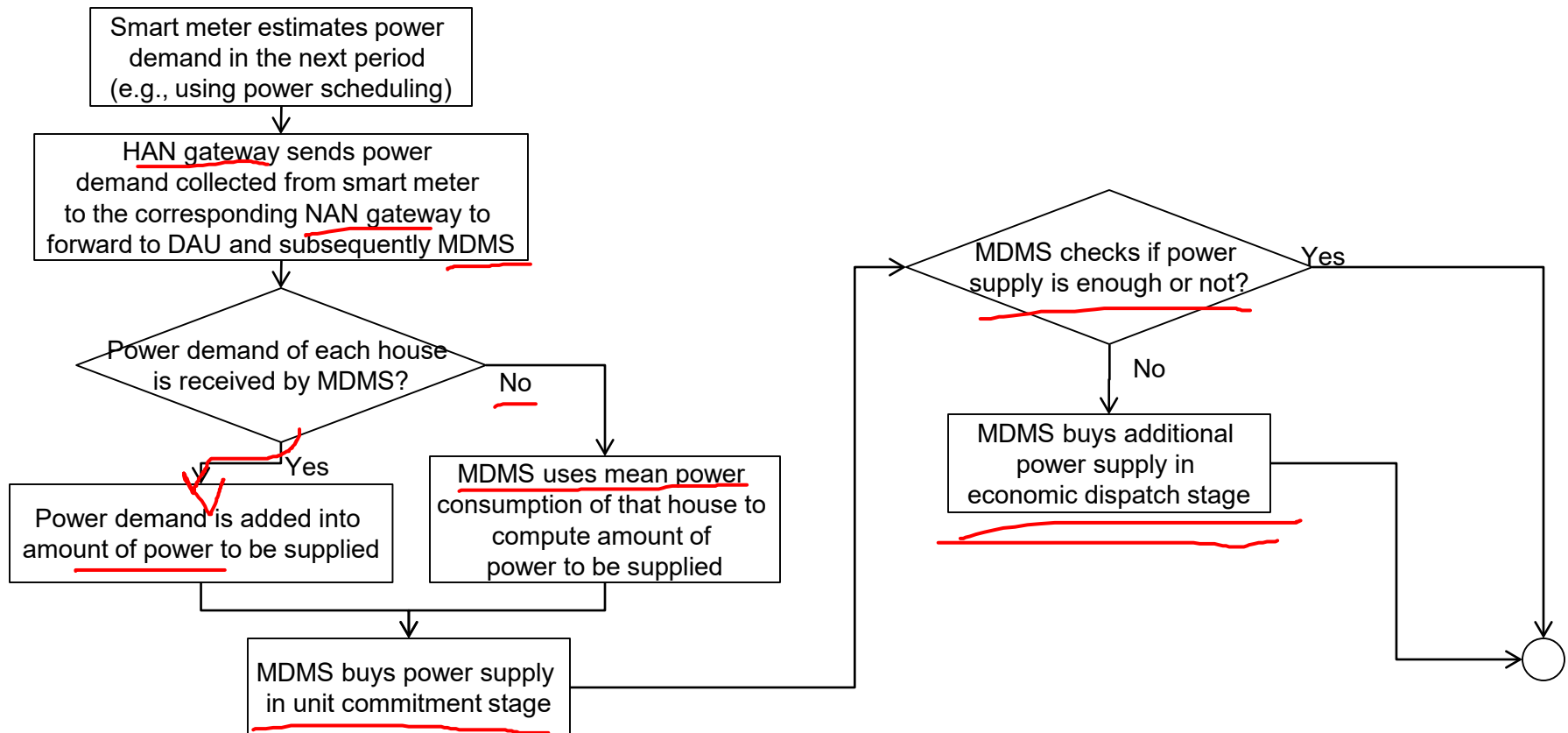
Reliability Analysis

- Reliability analysis of the wireless communications system in the smart grid can be performed
- Availability performance can be obtained given the random failure of the system devices
- Availability measure can be used to calculate the cost of power-demand estimation error and damage of power distribution equipment if its failure cannot be reported
- Redundancy design approaches can be developed to minimize the cost of failure as well as the cost of deployment of the wireless communications system in the smart grid



Advanced Metering Infrastructure

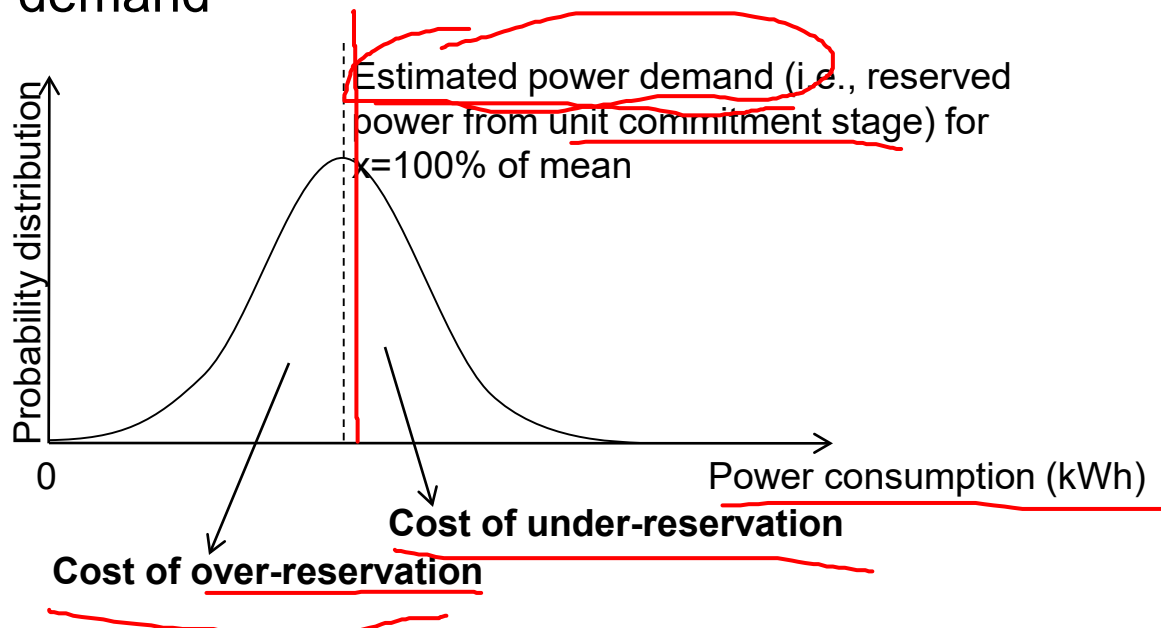
Reliability Analysis: Operation of a power system



Advanced Metering Infrastructure

Reliability Analysis: Operation of a power system

- If the power demand of any house is not received by the MDMS (e.g., due to failure of the HAN gateway, the NAN gateway, or the DAU), the MDMS uses historical data to compute the aggregated power demand
- $x\%$ of mean power-consumption¹ of those houses is used as the estimated demand



Advanced Metering Infrastructure

Reliability Analysis: Availability

- Availability of a component/device/system is the probability that the component/device/system has not failed or repaired and it can operate normally

$$A = \frac{\text{Uptime of component} \checkmark}{\text{Uptime of component} + \text{Downtime of component}}$$

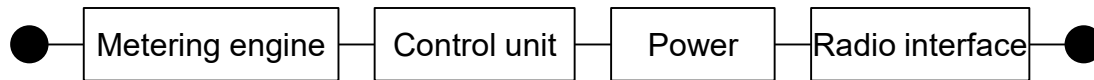
- Uptime is also known as the mean time between failure (MTBF)
- Downtime is known as the mean time between repair (MTBR)
- Failure rate can be obtained a 1-Availability



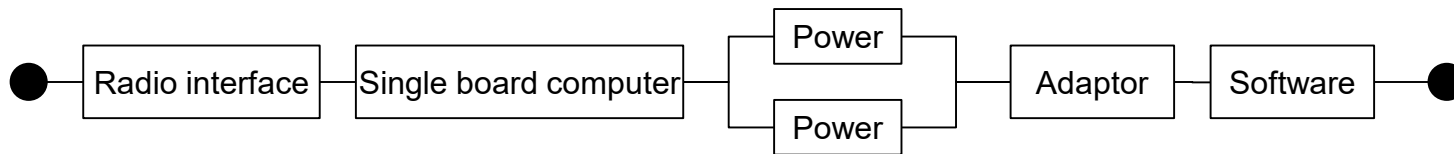
Advanced Metering Infrastructure

Reliability Analysis: Availability

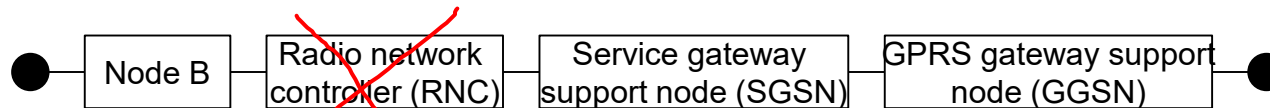
- Dependence diagram (DD) determines the contribution of each component to the availability of the system
- The components can be connected in parallel and/or series



Dependence diagram of smart meter and home area network gateway



Dependence diagram of neighborhood area network gateway



Dependence diagram of UMTS network

Advanced Metering Infrastructure

Reliability Analysis: Availability

- HAN gateway and a smart meter can be integrated into a single device. The availability of a HAN gateway is computed from $A_{HAN} = \text{availability of metering engine} \times \text{availability of control unit} \times \text{availability of power module} \times \text{availability of radio interface}$
- Availability of a NAN gateway is computed from $A_{NAN} = \text{availability of radio interface} \times \text{availability of single board computer} \times \text{availability of adaptor} \times \text{availability of software} \times (1 - (1 - \text{availability of power module})^2)$
- 3G cellular base station is assumed to have the DAU functionality whose availability is computed from: $A_{DAU} = \text{availability of node B} \times \text{availability of radio network controller (RNC)} \times \text{availability of service gateway support node (SGSN)} \times \text{availability of GPRS gateway support node (GGSN)}$



Summary

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